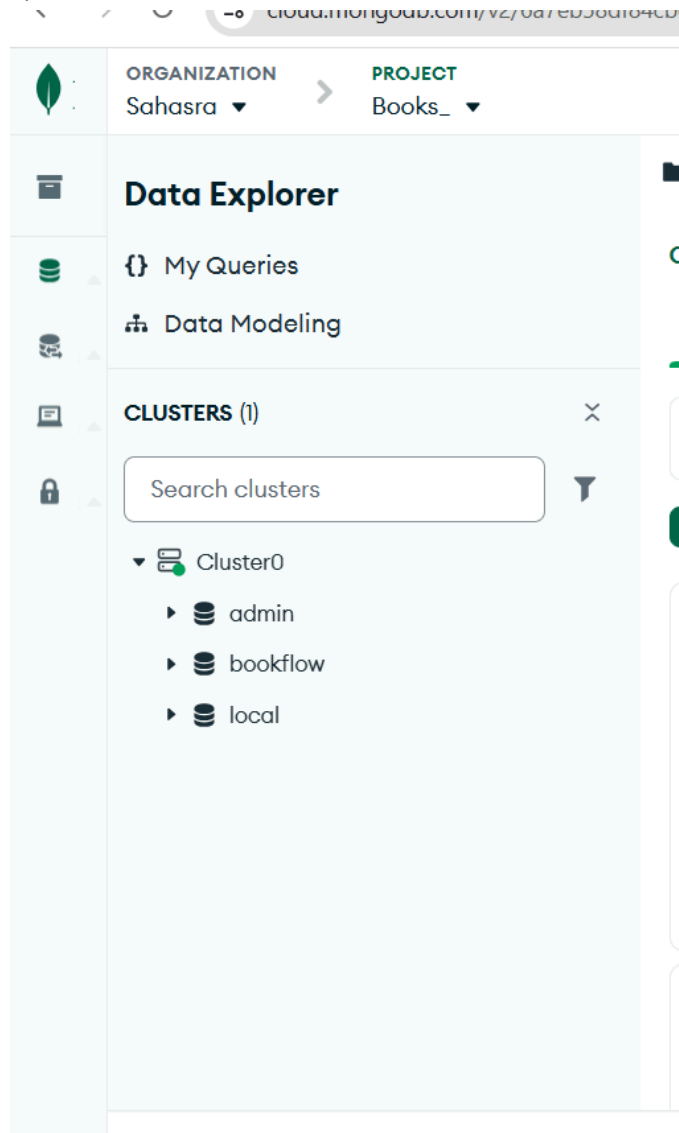


Week 3

Name: Sagi Sahasra Srihasa

Roll : 2520030008

1) Create a database bookflow



2) Create a new collection book_metadata

ORGANIZATION

Sahasra

PROJECT

Books_

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

bookflow

book_metadata

local

book_metadata

View monitor

Collection name

book_metadata

3) Click add data and insert documents



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PROJECT

Books_

📁

Data Explorer

🔍

My Queries

🔗

Data Modeling

CLUSTERS (1)

Cluster0

- admin
- bookflow
- book_metadata**
- local

book_metadata

bookflow

+

Cluster0

bookflow

book_metadata

Documents 5

Aggregations

Schema

Indexes

🕒

ADD DATA

BULK

EXPORT CODE

```

_id: ObjectId('6a7eb608f31ecc42859acf18')
book_id: 101
title: "The Martian"
author: "Andy Weir"
published_year: 2021
category: "Science Fiction"
format: "Digital PDF"
file_size: "2.5 MB"
member_id: 101
reviews: Array (3)

```


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CLUSTERS (1)

Search clusters

Cluster0

admin

bookflow

book_metadata

local

book_metadata

bookflow

+

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA

BULK

EXPORT CODE

format: "physical"

dimensions: Object (3)

member_id: 103

reviews: Array (2)

```
_id: ObjectId('6a7eb642f31ecc42859acf1e')
book_id: 104
title: "Artificial Intelligence Basics"
author: "John Smith"
published_year: 2023
category: "Artificial Intelligence"
format: "Digital PDF"
file_size: "4.2 MB"
member_id: 104
reviews: Array (3)
```

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Search clusters

Cluster0

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bookflow

book_metadata

local

book_metadata

bookflow

+

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA

BULK

EXPORT CODE

file_size: "4.2 MB"

member_id: 104

reviews: Array (3)

```
_id: ObjectId('6a7eb64ff31ecc42859acf20')
book_id: 105
title: "Data Structures"
author: "Mark Allen"
published_year: 2024
category: "Computer Science"
format: "Digital PDF"
file_size: "5.0 MB"
member_id: 105
reviews: Array (2)
```

4) Mongo DB Queries:

ORGANIZATION Sahasra PROJECT Books_ Compare tiers ?

Data Explorer

{} My Queries
Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bookflow
- book_metadata**
- local

book_metadata bookflow +

Cluster0 > bookflow > book_metadata View monitoring Visual

Documents 5 Aggregations Schema Indexes 1 Validation

```
{  
  "reviews.rating": {  
    "$gt": 4  
  }  
}
```

Generate query + Explain Reset Find

+ ADD DATA BULK EXPORT CODE 25 1 - 5 of 5

	file_size String	member_id Int32	reviews Array	dimensions Object
1	"2.5 MB"	101	[] 3 elements	No field
2	"3.1 MB"	102	[] 3 elements	No field
3	No field	103	[] 2 elements	{ } 3 fields
4	"4.2 MB"	104	[] 3 elements	No field
5	"5.0 MB"	105	[] 2 elements	No field

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{} My Queries
Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bookflow
- book_metadata**
- local

book_metadata bookflow +

Cluster0 > bookflow > book_metadata View r

Documents 5 Aggregations Schema Indexes 1 Validation

```
{  
  "reviews.rating": {  
    "$in": [4, 5]  
  }  
}
```

Generate query + Explain

+ ADD DATA BULK EXPORT CODE 25 1 - 5 of 5

	_id ObjectId	book_id Int32	title String	author Strin
1	ObjectId('6a7eb608f31ecc4...	101	"The Martian"	"Andy Weir"
2	ObjectId('6a7eb61af31ecc4...	102	"Dune"	"Frank Herbe
3	ObjectId('6a7eb627f31ecc4...	103	"Clean Code"	"Robert C. M
4	ObjectId('6a7eb642f31ecc4...	104	"Artificial Intelligence ...	"John Smith"
5	ObjectId('6a7eb64ff31ecc4...	105	"Data Structures"	"Mark Allen"

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CLUSTERS (1)

Search clusters

Cluster0

- admin
- bookflow
 - book_metadata**
 - local

book_metadatabookflow+

Cluster0 > bookflow > book_metadata

View monitoringVisualize Your Data

Documents 5AggregationsSchemaIndexes 1Validation

SmatchSunwindGenerate aggregation +? ExplainRunOptions

pipeline - modifiedSAVE + CREATE NEWEXPORT CODEPREVIEWSTAGESTEXTWIZARD

Stage 2Sunwind

1 "\$reviews"

Output preview after Sunwind stage (Sample of 10 documents)

```
_id: ObjectId('6a7eb608f31ecc42859acf18')
book_id: 101
title: "The Martian"
author: "Andy Weir"
published_year: 2021
category: "Science Fiction"
format: "Digital PDF"
file_size: "2.5 MB"
review_id: 101
```

```
_id: ObjectId('6a7eb608f31ecc42859acf18')
book_id: 101
title: "The Martian"
author: "Andy Weir"
published_year: 2021
category: "Science Fiction"
format: "Digital PDF"
file_size: "2.5 MB"
review_id: 101
```

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CLUSTERS (1)

Search clusters

Cluster0

- admin
- bookflow
 - book_metadata**
 - local

book_metadatabookflow+

Cluster0 > bookflow > book_metadata

View monitoringVisualize Your Data

Documents 5AggregationsSchemaIndexes 1Validation

SmatchSunwindSgroupGenerate aggregation +? ExplainRun

pipeline - modifiedSAVE + CREATE NEWEXPORT CODEPREVIEWSTAGESTEXT

Stage 3Sgroup

1 {
2 "_id": "\$title",
3 "average_rating": {
4 "\$avg": "\$reviews.rating"
5 },
6 "total_reviews": {
7 "\$sum": 1
8 }
9 }
10

Output preview after Sgroup stage (Sample of 4 documents)

```
_id: "The Martian"
average_rating: 4.666666666666667
total_reviews: 3
```

```
_id: "The Martian"
average_rating: 4.666666666666667
total_reviews: 3
```

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Search clusters

Cluster0

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bookflow

book_metadata

local

book_metadata

bookflow

+

Cluster0

bookflow

book_metadata

View monitoring

Visualize Your Data

Documents 5

Aggregations

Schema

Indexes 1

Validation

Smatch

Sunwind

Sgroup

Ssort

Generate aggregation

Explain

Run

Options

pipeline - modified

SAVE

CREATE NEW

EXPORT CODE

PREVIEW

STAGES

TEXT

WIZARD

Stage 4

Ssort

1

2

3

4

{

"average_rating": -1

}

Output preview after Ssort stage (Sample of 4 documents)

_id: "Dune"

average_rating: 4.666666666666667

total_reviews: 3

_id: "The Martian"

average_rating: 3.5

total_reviews: 1

6) Indexes:

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CLUSTERS (1)

Search clusters

Cluster0

admin

bookflow

book_metadata

local

book_metadata

bookflow

+

Cluster0

bookflow

book_metadata

View monitoring

Visualize Your Data

Documents 5

Aggregations

Schema

Indexes 2

Validation

Create Index

Refresh

Manage your search indexes

Name & Definition	Type	Size	Usage	Properties	Status
id	REGULAR	36.9 kB	8 (since Fri Aug 14 2026)	UNIQUE	READY
reviews.comments_text	TEXT	20.5 kB	0 (since Wed Aug 19 2026)		READY
_fts (text) _ftsx					

7) Search Using Text Index

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CLUSTERS (1)

Search clusters

Cluster0

admin

bookflow

book_metadata

local

book_metadata

bookflow

+

Cluster0

bookflow

book_metadata

View monitoring

Visualize Your Data

Documents 5

Aggregations

Schema

Indexes 2

Validation

Generate query

Explain

Reset

Find

Options

ADD DATA

BULK

EXPORT CODE

25

1-5 of 5

	book_metadata	_id ObjectId	book_id Int32	title String	author String	publ
1	ObjectId('6a7eb608f31ecc4...')	101		"The Martian"	"Andy Weir"	2021
2	ObjectId('6a7eb627f31ecc4...')	103		"Clean Code"	"Robert C. Martin"	2019
3	ObjectId('6a7eb61af31ecc4...')	102		"Dune"	"Frank Herbert"	2022
4	ObjectId('6a7eb64ff31ecc4...')	105		"Data Structures"	"Mark Allen"	2024
5	ObjectId('6a7eb642f31ecc4...')	104		"Artificial Intelligence"	"John Smith"	2023

8) Search for a specific keyword

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Books_

Compare tiers

95

Data Explorer

{ } My Queries

⚙ Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

bookflow

book_metadata

local

book_metadata

bookflow

+

Cluster0 > bookflow > book_metadata

View monitoring

Visualize Your Data

Documents 5

Aggregations

Schema

Indexes 2

Validation

⌵

{

"\$text": {

"\$search": "\"science fiction\""

}

}

}

Generate query

Explain

Reset

Find

Options

ADD DATA

BULK

EXPORT CODE

25

1 - 2 of 2

book_metadata

	_id ObjectId	book_id Int32	title String	author String	publ
1	ObjectId('6a7eb61af31ecc4...	102	"Dune"	"Frank Herbert"	2022
2	ObjectId('6a7eb608f31ecc4...	101	"The Martian"	"Andy Weir"	2021